

SLAY THE ROADMAP

Slay The Roadmap

Teach the basics of computer programming using a gamified approach based on structured learning roadmaps.

Course:

Engineering Gamified Systems (2025/2026)

Team Members:

Agostino D'Agostino
Alessandro Di Giacomo

Github Prototype:

https://github.com/Ago95Dev/slay_the_roadmap

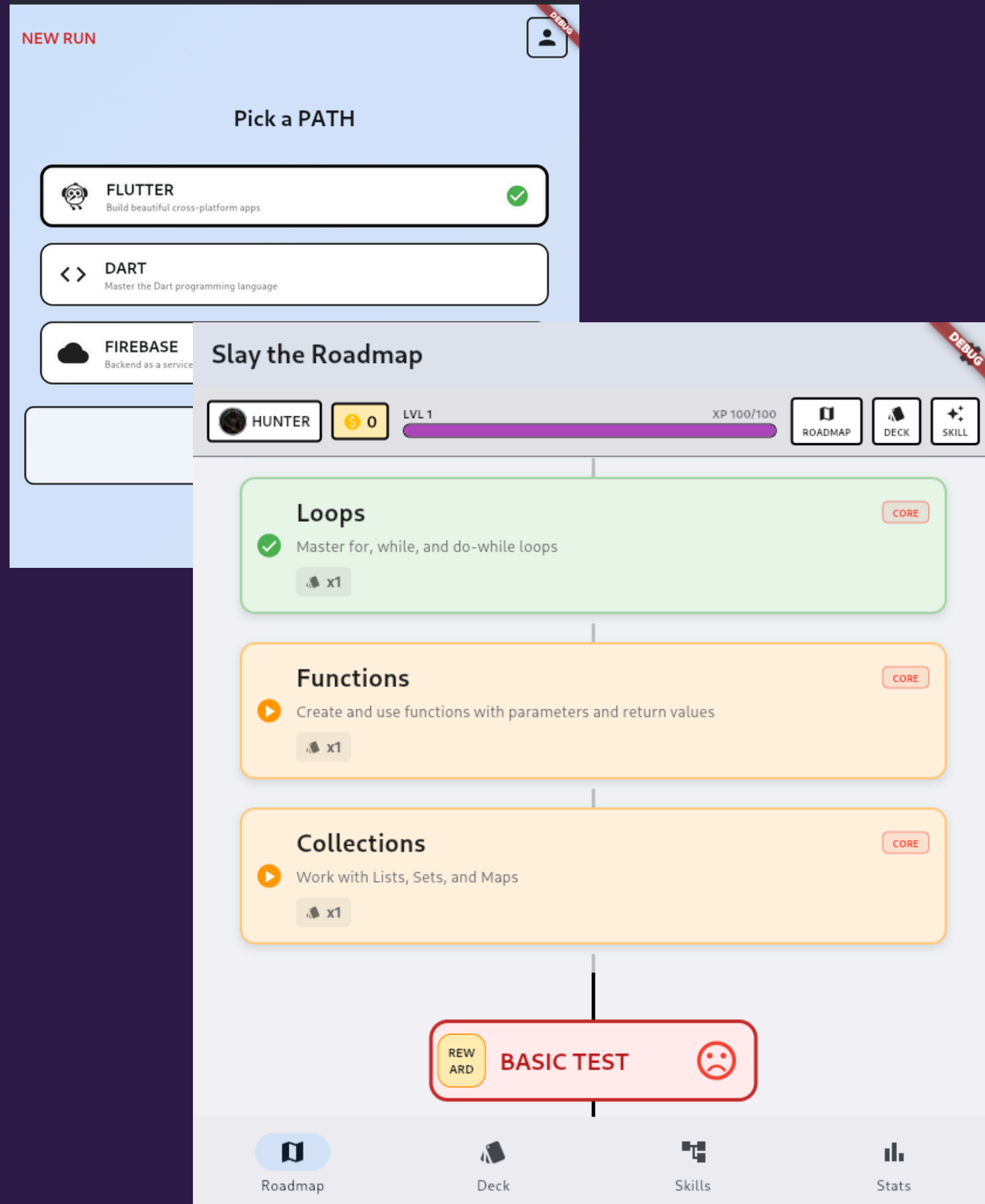
Purpose, Objectives & Target Users

- * **Problem:** Traditional programming apps for education often suffers from a steep learning curve and a lack of immediate feedback.

This gap frequently results in low engagement and high attrition rates among learners.

- * **Solution:**
 - Teach the fundamentals of computer programming through a gamified approach based on structured learning paths.
 - Facilitate the acquisition of skills by integrating playful engagement mechanics, making learning fun and consistent.

- * **Target Users:**
 - Intended for a user base ranging from absolute beginners (7+ years) to developers approaching a new language.
 - Covers both self-learning contexts and institutional training.



Core Concept & Gamification Strategy

Core Idea: An Educational Deck-Building Roguelike.

The Roadmap is the "Dungeon". Topics are the "Monsters". Concepts are the "Cards".

The Loop:

- 1 **Preparation:** Study the learning materials.
- 2 **Drafting:** Build your deck with acquired knowledge.
- 3 **The Encounter:** Face the Boss (Exam).

User Experience & Usability

Structured User Journey:

Study Phase: Focused reading of educational materials for knowledge acquisition.

Quiz Phase: Verification step requiring a >80% pass rate to proceed.

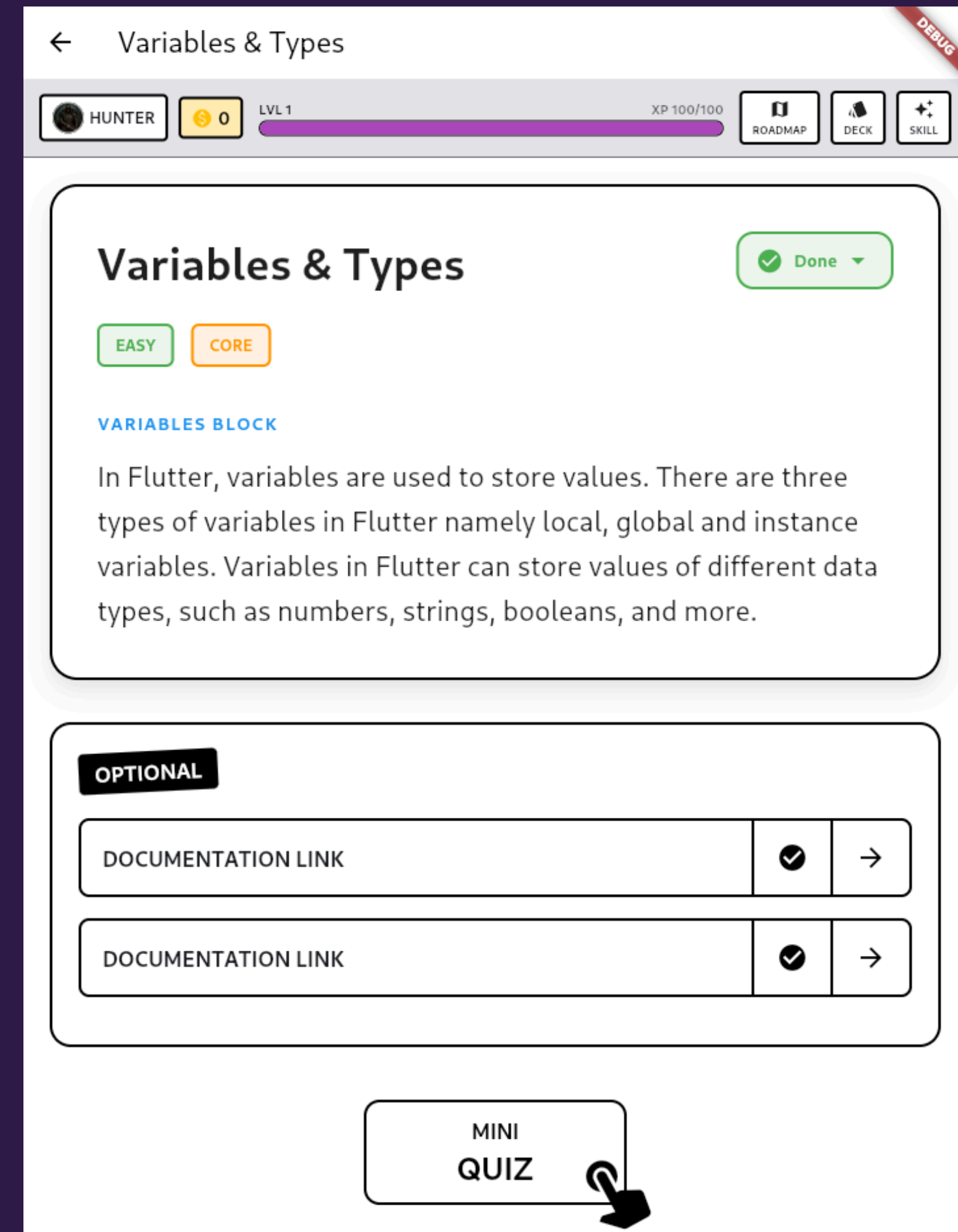
Deck Building: Strategic reward selection and deck optimization.

Boss Fight: Practical skill check via turn-based combat.

UX Design Choices:

Cross-Platform Compatibility: Developed using the Flutter framework to ensure a unified experience across mobile, web, and desktop via a single codebase.

Audio-Visual Feedback: Context-differentiated soundtracks and sound effects for study and combat phases.



Feedback, Progression & Rewards

← Variables & Types

DEBUG

HUNTER

\$ 0

LVL 1

XP 100/100

ROADMAP

DECK

SKILL

Question 1 of 5

Which keyword is used to declare a variable that can be reassigned?

var

final

const

let

✓ Explanation

var is used for variables that can be reassigned. Use final for variables that can only be set once.

Next Question

Progression:

The progression is organized into thematic chapters of 5-7 topics each, with a final boss representing the summative test.

Rewards:

An experience system (XP) rewards the completion of tasks with level progression and scalar rewards. Cards collection and evolution introduces a layer of improvement to the experience.

Feedbacks:

Users receive immediate visual cues and notifications for every answer, whether correct or incorrect. This mechanism is designed to improve learning at every step, providing real-time validation that guides the player towards learning the material.

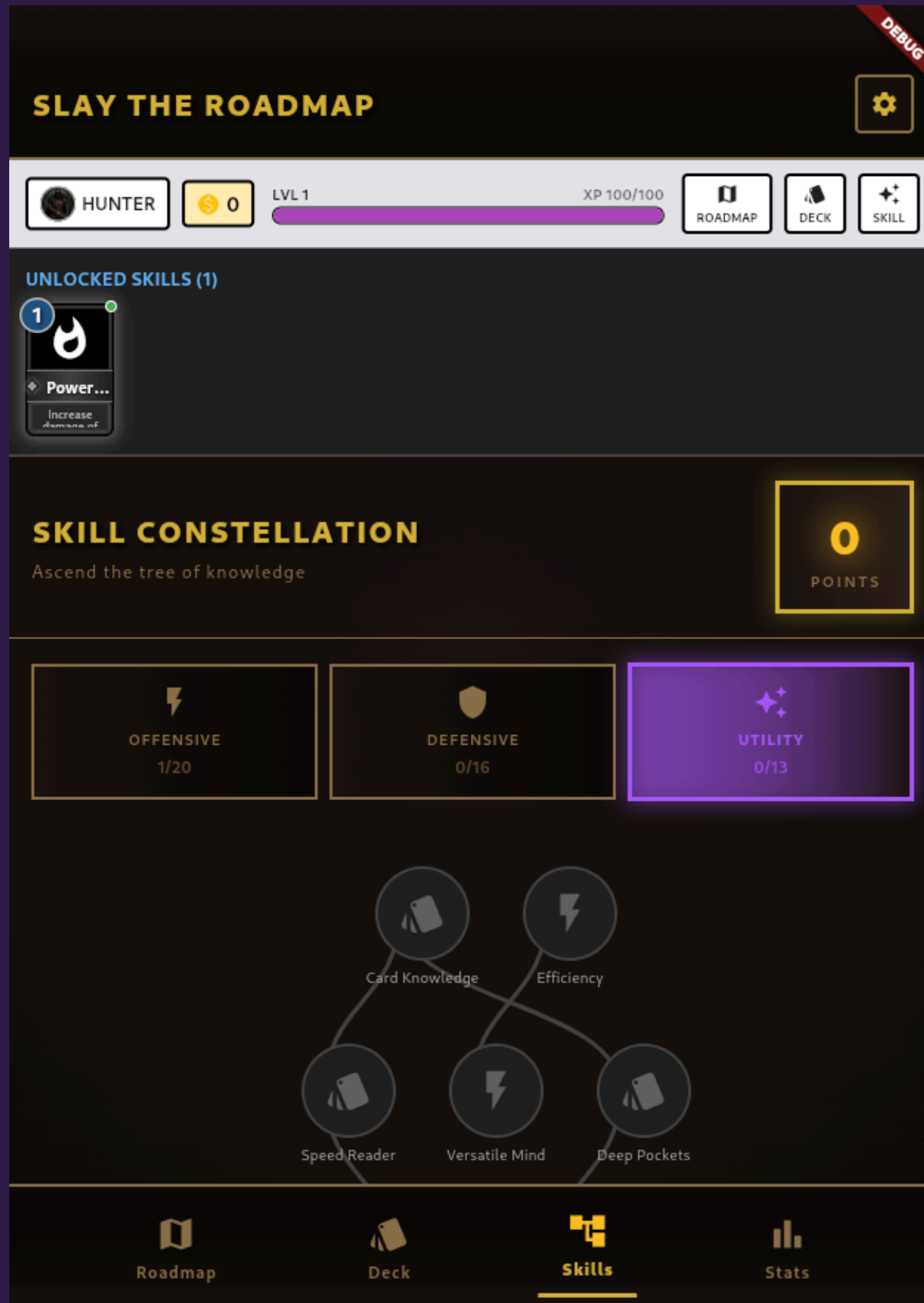
Real-World Use, Reusability & Scalability

Reusability:

While our current prototype focuses on computer programming, the underlying architecture can be adapted to any content. The application structure can be used to teach any topic.

Scalability:

This flexibility makes Slay The Roadmap highly scalable and suitable for diverse real-world contexts, including autonomous self-learning platforms where standardized, rapid skill acquisition is required.



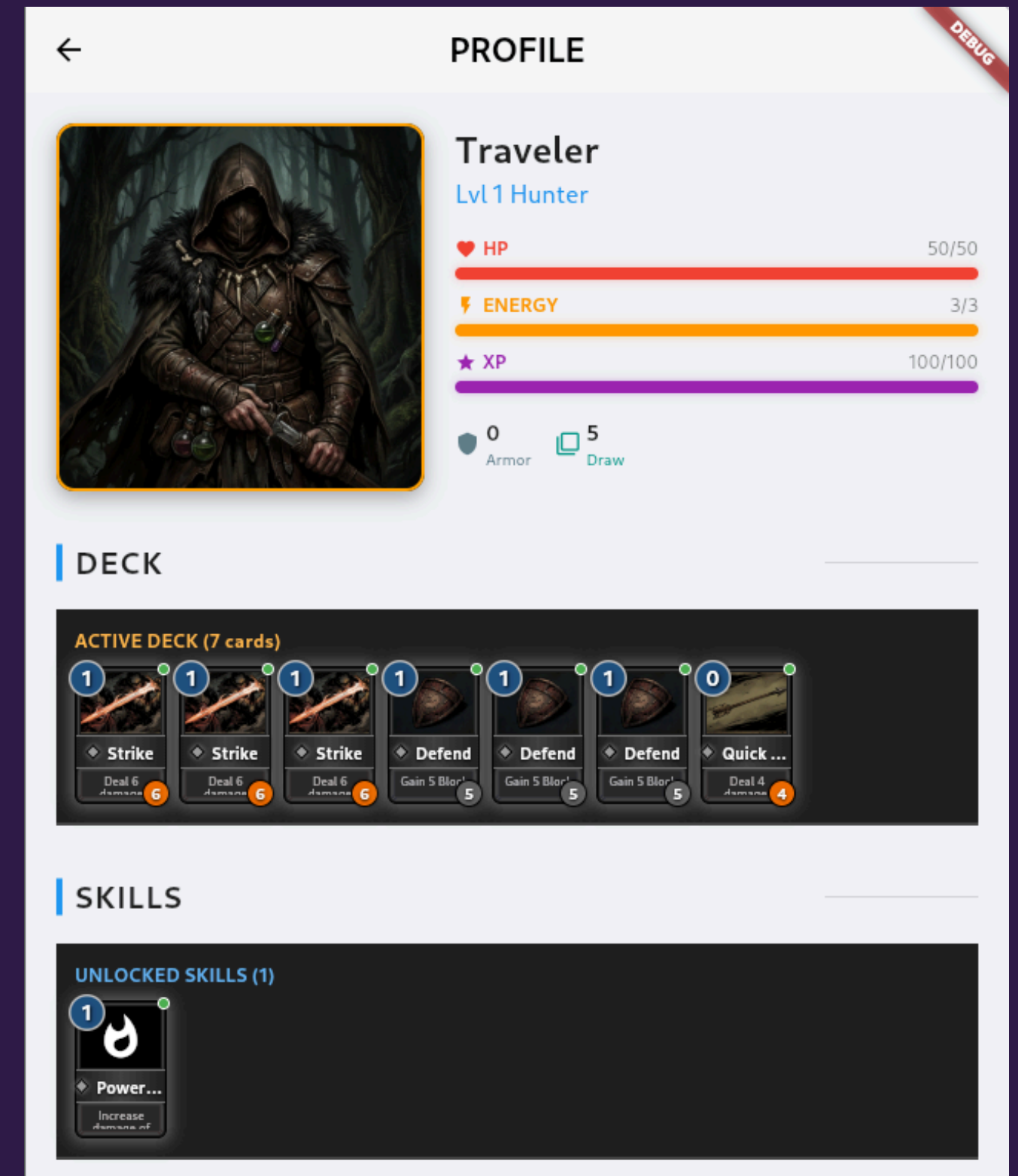
Motivation & Engagement Over Time

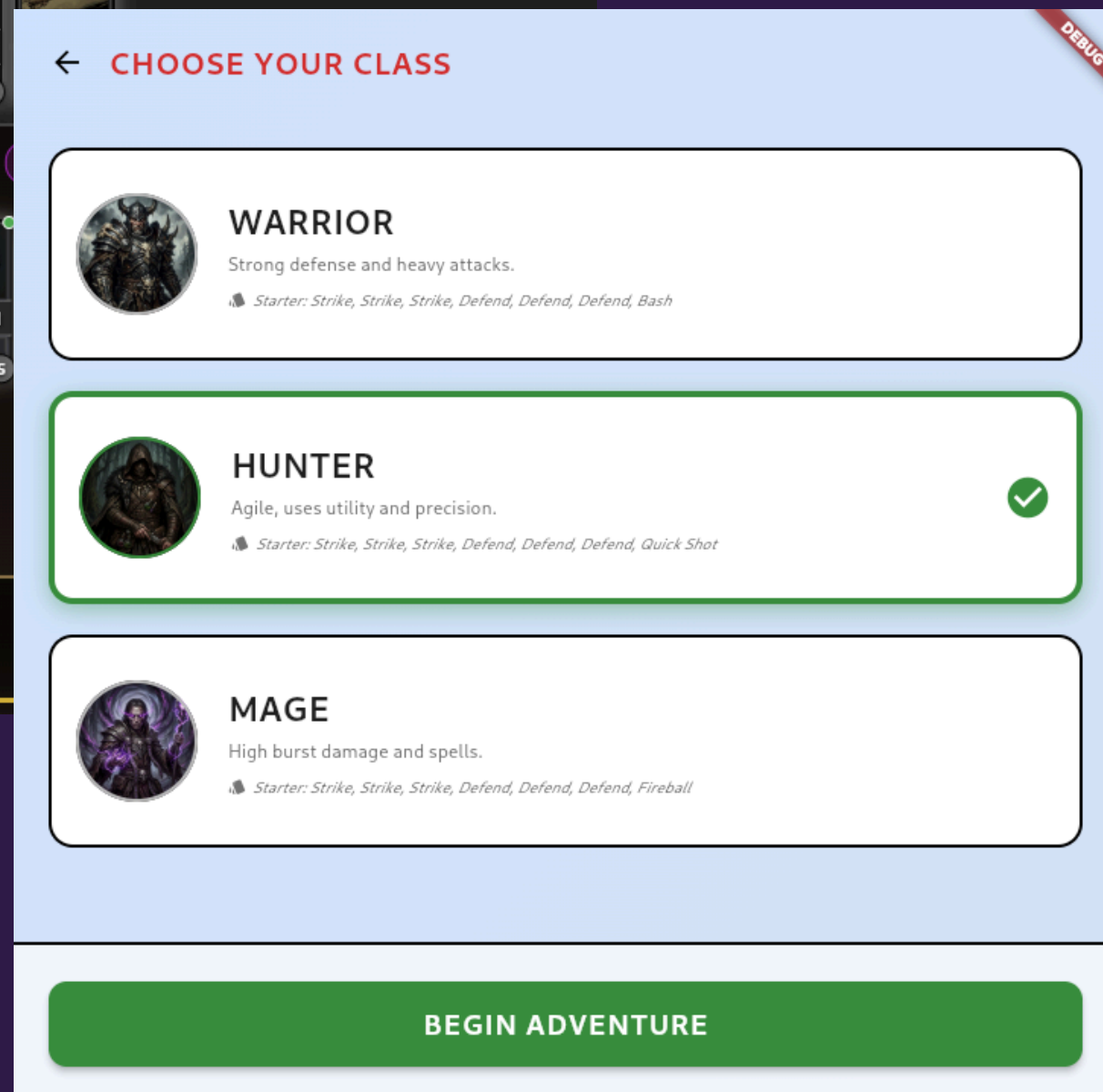
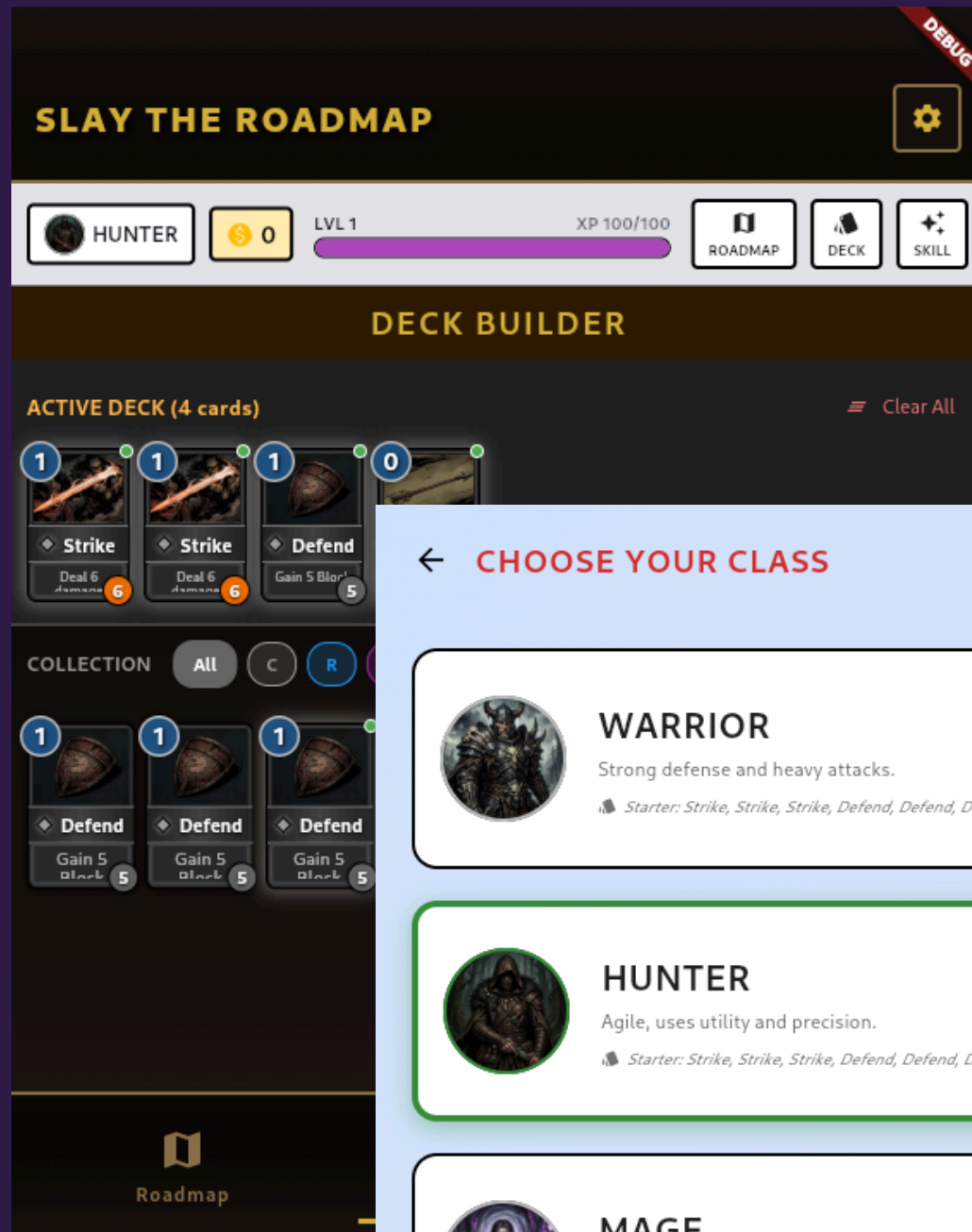
Motivation & Engagement:

We employ a dual strategy. In the short term, motivation is driven by risk/reward decisions presented through narrative events, requiring players to carefully evaluate the consequences of their resource usage.

In the long term, the experience shifts towards strategic depth. Through the Skill Tree and card collection mechanics, players engage in "theory-crafting," optimizing their builds to suit their playstyle.

Additionally, the aesthetic personalization of user profiles allows students to express their identity, fostering a deeper connection with the platform.





Originality & Technological Quality

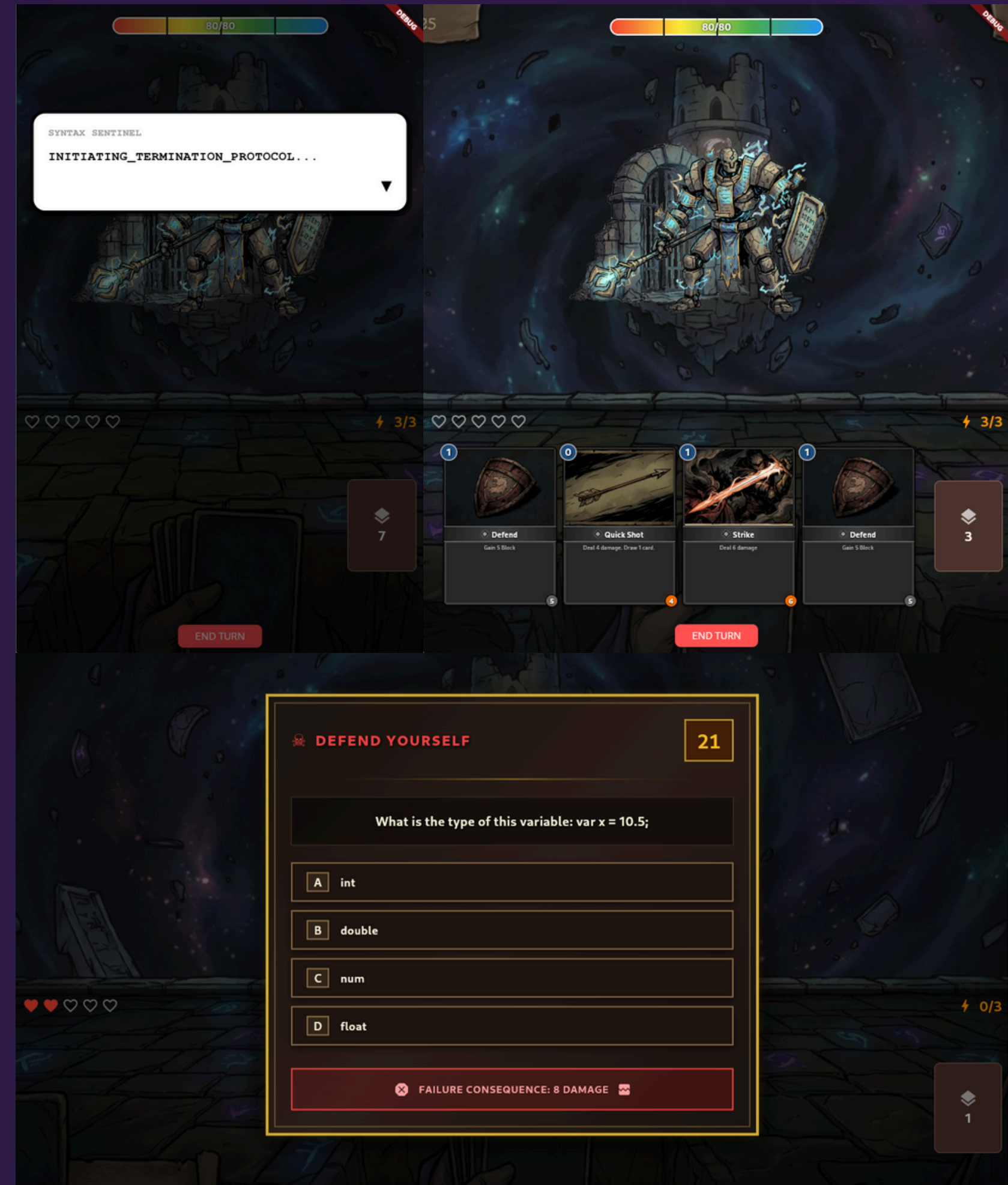
Originality: What sets Slay The Roadmap apart from standard ed-tech tools is the shift from passive recall to active strategic optimization. Unlike simple quizzes, our system demands contextualized problem-solving: the player isn't just answering a question, they are analyzing a problem (the Boss) and custom-tailoring a solution (the Deck) to exploit specific weaknesses.

Technologically: this depth is supported by a robust architecture built on Flutter, which handles complex state management for turn-based logic and randomized card interactions. This ensures a polished experience where the difficulty dynamically adapts to the user's skill level, maintaining the delicate balance of the "Flow" state.

Theme Alignment & Impact

Theme alignment: We adopted a Minimal Dark Fantasy aesthetic not merely for style, but for function. It reduces visual fatigue during long study sessions and creates an immersive atmosphere that reframes the learning experience.

Impact: The educational impact is significant: we aim to replace exam anxiety with the excitement of an epic battle. By contextualizing abstract technical concepts into tangible entities we aid memorization and make trial and error an accepted, integral part of the adventure.



Demo, Strengths, Limits & Roadmap



Strengths: Our system's primary strength is the Adaptive Flow: the game dynamically adjusts challenges to prevent boredom or frustration. Furthermore, the Dual Progression System ensures that as the in-game avatar grows in power, the user grows in actual technical competence, creating a perfect synergy between gameplay and education.

Limits & Mitigation: We acknowledge potential barriers such as Quiz Anxiety, which we mitigate by reframing errors as "battle damage" rather than academic failure, and providing unlimited attempts. Additionally, Competitive Stress is managed by making leaderboards strictly optional, prioritizing individual growth over comparison.

Roadmap: Looking ahead, our roadmap includes the development of a "Boss Creator" tool and the introduction of Knowledge Cards: special rewards earned by mastering specific topics that allow players to influence and diversify future Boss questions, adding a strategic layer to how they diversify their learning path.

Thank You!